

UAPOML Summer Project: Week 1 Assignment

Theoretical Questions

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Question 1

Given Prices: 100, 108, 103, 115, 110.

(a) **Simple Returns:** $R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$

- Month 1: $(108 - 100)/100 = 0.0800$ (8.00%)
- Month 2: $(103 - 108)/108 = -0.0463$ (-4.63%)
- Month 3: $(115 - 103)/103 = 0.1165$ (11.65%)
- Month 4: $(110 - 115)/115 = -0.0435$ (-4.35%)

(b) **Log Returns:** $r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$

- Month 1: $\ln(108/100) = 0.0770$ (7.70%)
- Month 2: $\ln(103/108) = -0.0474$ (-4.74%)
- Month 3: $\ln(115/103) = 0.1102$ (11.02%)
- Month 4: $\ln(110/115) = -0.0445$ (-4.45%)

(c) **Approximation:** The approximation $r_t \approx R_t$ holds as $0.0770 \approx 0.0800$, $-0.0474 \approx -0.0463$, $0.1102 \approx 0.1165$, and $-0.0445 \approx -0.0435$. The approximation is **least accurate at Month 3**. This occurs because the absolute magnitude of the return is the highest (11.65%). The Taylor series expansion of $\ln(1+x) \approx x$ becomes progressively less accurate as the absolute value of x moves further away from zero.

Question 2

(a) Linearity of expectation states that the expected value of a sum of random variables is the sum of their expected values, scaled by their constants: $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$. Applying this to the portfolio:

$$\begin{aligned}\mathbb{E}[R_p] &= w_1\mathbb{E}[R_1] + w_2\mathbb{E}[R_2] \\ &= 0.4(0.15) + 0.6(0.07) \\ &= 0.06 + 0.042 = 0.102 \text{ (or 10.2\%)}\end{aligned}$$

(b) No, this result does not change. The linearity of expectation holds universally, regardless of whether the random variables (the asset returns) are independent or correlated.

Question 3

Starting with the definition of variance for the sum of two random variables:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

Let $X = w_1 R_1$ and $Y = w_2 R_2$. Substituting these into the formula yields:

$$\text{Var}(w_1 R_1 + w_2 R_2) = \text{Var}(w_1 R_1) + \text{Var}(w_2 R_2) + 2\text{Cov}(w_1 R_1, w_2 R_2)$$

Using the properties of variance ($\text{Var}(aX) = a^2 \text{Var}(X)$) and covariance ($\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$), we get:

$$\sigma_p^2 = w_1^2 \text{Var}(R_1) + w_2^2 \text{Var}(R_2) + 2w_1 w_2 \text{Cov}(R_1, R_2)$$

Substituting standard notation $\text{Var}(R_i) = \sigma_i^2$ and the definition of covariance $\text{Cov}(R_1, R_2) = \rho_{12} \sigma_1 \sigma_2$:

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

Question 4

Given: $\sigma_A = 0.25$, $\sigma_B = 0.12$, $w_A = 0.5$, $w_B = 0.5$. The variance formula reduces to:

$$\begin{aligned} \sigma_p^2 &= (0.5)^2 (0.25)^2 + (0.5)^2 (0.12)^2 + 2(0.5)(0.5)\rho(0.25)(0.12) \\ &= 0.015625 + 0.0036 + 0.015\rho \\ &= 0.019225 + 0.015\rho \end{aligned}$$

(a)

- $\rho = 1$: $\sigma_p^2 = 0.019225 + 0.015(1) = 0.034225 \implies \sigma_p = 0.185$ (18.5%)
- $\rho = 0$: $\sigma_p^2 = 0.019225 + 0.015(0) = 0.019225 \implies \sigma_p \approx 0.1387$ (13.87%)
- $\rho = -1$: $\sigma_p^2 = 0.019225 + 0.015(-1) = 0.004225 \implies \sigma_p = 0.065$ (6.5%)

(b) It equals the weighted average when $\rho = +1$. Algebraic proof:

$$\begin{aligned} \sigma_p^2 &= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B (1) \sigma_A \sigma_B \\ \sigma_p^2 &= (w_A \sigma_A + w_B \sigma_B)^2 \\ \sigma_p &= w_A \sigma_A + w_B \sigma_B \end{aligned}$$

(c) For $\rho = -1$, risk is zero when $w_A \sigma_A - (1 - w_A) \sigma_B = 0$.

$$\begin{aligned} 0.25w_A - 0.12(1 - w_A) &= 0 \\ 0.25w_A - 0.12 + 0.12w_A &= 0 \\ 0.37w_A &= 0.12 \implies w_A^* = \frac{12}{37} \approx 0.3243 \end{aligned}$$

Question 5

When two assets are combined, portfolio variance is determined by $\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho\sigma_1\sigma_2$. If correlation $\rho < 1$, the covariance term is strictly smaller than $2w_1w_2\sigma_1\sigma_2$. Therefore, the total variance σ_p^2 evaluates to less than $(w_1\sigma_1 + w_2\sigma_2)^2$, which is the squared weighted average of individual risks. Economically, imperfectly correlated assets do not move in perfect lockstep. When one asset underperforms, the other is likely to partially offset that loss. This structural dampening of combined volatility creates a diversification benefit, lowering overall portfolio risk below the simple weighted sum.

Question 6

(a) Calculating the covariance matrix elements using $\Sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$:

- $\Sigma_{11} = (0.25)^2 = 0.0625$
- $\Sigma_{22} = (0.12)^2 = 0.0144$
- $\Sigma_{33} = (0.04)^2 = 0.0016$
- $\Sigma_{12} = \Sigma_{21} = 0.4(0.25)(0.12) = 0.0120$
- $\Sigma_{13} = \Sigma_{31} = 0.1(0.25)(0.04) = 0.0010$
- $\Sigma_{23} = \Sigma_{32} = 0.2(0.12)(0.04) = 0.00096$

$$\Sigma = \begin{bmatrix} 0.0625 & 0.0120 & 0.0010 \\ 0.0120 & 0.0144 & 0.00096 \\ 0.0010 & 0.00096 & 0.0016 \end{bmatrix}$$

(b) The matrix is symmetric because covariance is commutative ($\text{Cov}(A, B) = \text{Cov}(B, A)$). Visually, the off-diagonal elements satisfy $\Sigma_{ij} = \Sigma_{ji}$ across the main diagonal.

Question 7

(a) Calculating the Sharpe Ratio ($SR = \frac{\mathbb{E}[R_p] - R_f}{\sigma_p}$):

- Portfolio X: $(0.14 - 0.04)/0.22 = 0.4545$
- Portfolio Y: $(0.09 - 0.04)/0.10 = 0.5000$
- Portfolio Z: $(0.20 - 0.04)/0.35 = 0.4571$

(b) Ranking: $Y > Z > X$. A rational mean-variance investor prefers Portfolio Y because it has the highest Sharpe Ratio, indicating it delivers the greatest excess return per unit of risk taken.

(c) Even though Portfolio Z has the highest absolute return, a rational investor can borrow cash at the risk-free rate to leverage their investment in Portfolio Y. This leveraged position in Y can match Z's 20% expected return while still maintaining a lower overall standard deviation than 35%.

Question 8

(a) The Global Minimum Variance (GMV) portfolio is the combination of risky assets that produces the absolute lowest possible variance. It is the leftmost point on the minimum variance frontier because the horizontal axis represents risk (σ); no portfolio can geometrically exist to the left of the lowest possible risk boundary.

(b) The student is incorrect. The lower half of the minimum variance frontier is inefficient. For every portfolio on the lower half, there exists a corresponding portfolio exactly above it on the upper half that holds identical risk but offers a higher expected return. A rational investor would exclusively hold portfolios on the upper half.

(c) The Capital Market Line (CML) is constructed by drawing a straight tangent line from the risk-free rate on the y-axis to the efficient frontier of risky assets. The slope of the CML represents the Sharpe Ratio of the tangency portfolio (which is the optimal risky market portfolio).

Question 9

(a) The standard error of the sample mean is given by $SE = \frac{\sigma}{\sqrt{n}}$.

$$SE = \frac{0.30}{\sqrt{36}} = \frac{0.30}{6} = 0.05 \text{ (or 5\%)}$$

This implies that relying on historical data alone is highly unreliable; a 5% margin of error is massive compared to standard monthly expected returns, meaning the historical average is largely statistical noise.

(b) Mean-Variance Optimisation acts as an "error maximiser" because it aggressively over-allocates capital to assets with the highest estimated returns and lowest estimated variances, which are precisely the inputs most likely to contain positive estimation errors (i.e., historical outliers due to luck). Machine learning mitigates this by identifying structural predictive features rather than relying purely on noisy historical averages, allowing for better regularisation and out-of-sample robustness.

(c) An $N \times N$ covariance matrix has $\frac{N(N+1)}{2}$ unique parameters. For $N = 50$, this requires estimating $\frac{50(51)}{2} = 1,275$ unique parameters. This is a significant challenge because accurately estimating 1,275 interacting variables requires a prohibitively large volume of clean, stationary historical data; otherwise, the resulting matrix will be dominated by noise and mathematically unstable.